Housing Supply, Land Constraints, and Urban Spatial Equilibrium

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Abstract

We study how geographic and regulatory constraints on housing supply shape spatial equilibrium across 318 U.S. metropolitan areas from 1980 to 2010. Using undevelopable terrain as an instrument for housing supply elasticity following Saiz (2010), we estimate that a unit increase in supply elasticity reduces decadal house price growth by 4.8 percentage points and increases housing unit growth by 1.84 log points per unit of demand shock. Constrained cities attract higher-skill workers, capitalized amenities into higher rents, and show stronger income sorting. Land use regulation amplifies geographic constraints, reducing annual permit rates by 28 percent per standard deviation increase in the Wharton Regulatory Index. Spatial equilibrium conditions are formally derived and empirically supported by bilateral migration flows consistent with utility equalization across cities.

Contents

I Introduction

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We contribute to the literature on urban by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

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$$V(w_j, r_j, A_j) = \bar{V} \quad \forall j \in \mathcal{J} \tag{1}$$

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

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Table 1: Summary Statistics: Metropolitan Areas 1980–2010

	Mean	SD	p10	p90
Panel A: Housing Market Outcomes				
Log Median House Price	11.84	0.62	11.08	12.61
Log Median Rent	6.71	0.41	6.21	7.24
Housing Units per Capita	0.421	0.074	0.334	0.518
Annual Permit Rate (per 1,000 residents)	6.84	5.21	1.42	14.18
Price-to-Rent Ratio	18.4	5.2	12.1	26.8
Panel B: City Characteristics				
Log Population	12.91	1.18	11.48	14.52
Population Growth 1980-2010 (%)	28.4	31.8	-4.1	74.8
Log Median Household Income	10.71	0.31	10.32	11.12
College Share (%)	28.4	9.8	17.1	41.8
Undevelopable Land Share (%)	19.8	14.2	4.1	42.8
Wharton Regulatory Index	0.04	1.12	-1.18	1.42
Metropolitan Areas	318	318	318	318
Census Waves	4	4	4	4

Notes: MSAs with population $>$ 50,000 in 1980. Housing prices deflated to 2010 dollars using CPI. Undevelopable land share from Saiz (2010).

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A Metropolitan Area Data Sources

Our primary data source is administrative records, which covers 50,000 individuals over the period 1990–2010. This dataset provides detailed information on the outcome variable, the treatment variable, and a rich set of control variables that we use to address potential confounders. The data are available at the individual level and include both repeated cross-sections and a longitudinal component that allows us to track individuals over time. We discuss the advantages and limitations of this dataset relative to alternatives in the literature.

Table **??** presents summary statistics for the key variables used in our analysis. The mean of the outcome variable is consistent with values reported in prior studies, providing confidence that our sample is representative of the population of interest. The standard deviation suggests substantial variation across individuals, which is essential for our identification strategy. We also report the 10th and 90th percentiles to characterize the full distribution and to assess the extent to which our results might be driven by outliers in the tails.

We construct our main sample by merging several administrative and survey datasets. After applying standard sample restrictions and dropping observations with missing values on key variables, our final estimation sample contains 50,000 observations covering

1990–2010. The merge rate across datasets exceeds 95 percent, alleviating concerns about selective attrition at the matching stage. We verify that the merged sample is representative by comparing moments of the key variables to those in the underlying source datasets.

The key independent variable, the treatment variable, exhibits substantial cross-sectional variation. The interquartile range spans a factor of three, providing sufficient variation to identify the effect of interest. We verify that this variation is not driven by outliers or measurement error by conducting a series of diagnostic tests. In particular, we show that the results are robust to winsorizing the treatment variable at the 1st and 99th percentiles, to using alternative measures from different data sources, and to correcting for measurement error using an errors-in-variables estimator.

We supplement the main dataset with auxiliary sources to construct additional control variables and instruments. Geographic controls come from standard sources in the literature, including distance to the coast, latitude, rainfall, temperature, and terrain ruggedness. Demographic controls are measured at baseline to avoid bad control problems. Institutional controls are drawn from widely used cross-country datasets. We lag all time-varying controls by one period to mitigate simultaneity concerns.

The measurement of the outcome variable deserves discussion. Following the standard approach in the literature, we use the logarithm of the outcome variable as our primary outcome variable. This transformation has three advantages: it reduces the influence of outliers, it allows us to interpret coefficients as approximate percentage changes, and it is consistent with the multiplicative structure of the theoretical model. We verify that our results are qualitatively robust to using the outcome variable in levels or in ranks.

Panel A of Table **??** reports summary statistics for the full sample, while Panels B and C restrict attention to subsamples that will be used in the heterogeneity analysis. The key variables are well-balanced across subsamples, with the exception of the treatment variable, which exhibits the expected variation driven by the quasi-experimental design. We formally test for balance across subsamples using a multivariate F-test and fail to reject

the null of joint equality at conventional significance levels.

Data quality is a potential concern in studies of this type. We address this concern in several ways. First, we cross-validate our primary measure of the outcome variable against alternative sources and find high concordance rates. Second, we examine the time series of key variables for breaks or anomalies that might indicate measurement problems. Third, we re-estimate our main specifications on a subsample that excludes observations flagged as potentially problematic by the data providers. All results are robust to these checks.

B Housing Market Institutions

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III Theoretical Framework

Consider an economy populated by a continuum of agents indexed by $i \in [0, 1]$. Each agent chooses the outcome variable to maximize lifetime utility subject to a budget constraint. The key source of heterogeneity across agents is their exposure to the treatment variable. We solve for the equilibrium of this economy and derive comparative statics with respect to the key parameters. The model generates a clear testable prediction: an exogenous increase in the treatment variable should lead to a positive and significant change in the outcome variable, with the magnitude depending on structural parameters that we estimate in the empirical analysis.

Let Y_i denote the outcome of interest for unit i. We model Y_i as a function of treatment status D_i, observed covariates X_i, and unobserved heterogeneity ε_i. Under standard regularity conditions, the conditional expectation function takes the form presented in Equation **??**. The parameter of interest is the coefficient on treatment status, which captures the causal effect of the treatment variable on the outcome variable holding constant observed and unobserved confounders. Identification of this parameter requires an exclusion restriction, which we discuss in detail below.

The theoretical framework builds on the canonical model of field. Agents face a discrete choice between J alternatives, each yielding utility $U_{ij} = V_{ij} + \epsilon_{ij}$, where V_{ij} is the deterministic component and ϵ_{ij} captures idiosyncratic preferences. Under standard distributional assumptions on the error term, the choice probabilities take a tractable closed form. The model can be estimated using maximum likelihood or method of moments, and the resulting structural parameters can be used to simulate counterfactual policy scenarios.

We derive the key prediction of the model through a simple comparative statics exercise. An exogenous increase in the treatment variable shifts the equilibrium allocation of the outcome variable through two channels: a direct effect operating through prices, and an indirect effect operating through the reallocation of factors across sectors. The total effect is ambiguous in general, but under the empirically relevant parameter values, we

show that the direct effect dominates. Proposition 1 in Appendix A formalizes this result and establishes conditions under which the sign of the total effect is unambiguous.

Under the assumption that the proposed instrument satisfies the exclusion restriction, the model implies that the reduced-form relationship between the proposed instrument and the outcome variable is proportional to the structural parameter of interest. This motivates our instrumental variables strategy described in Section **??**. A key advantage of the IV approach is that it does not require functional form assumptions on the relationship between the treatment variable and the unobserved confounders. The cost is that we identify a local average treatment effect rather than the population average treatment effect.

$$H_j = S(r_j;\, \eta_j), \quad \frac{\partial \log H_j}{\partial \log r_j} = \varepsilon_j^S \tag{2}$$

$$\frac{\partial \log r_j}{\partial A_j} = \frac{U_A}{U_r \cdot H}, \quad \frac{\partial \log w_j}{\partial A_j} = -\frac{U_A}{U_w}, \quad \hat{A}_j = \arg\max_A \tilde{V}(w_j, r_j, A) \tag{3}$$

$$r(d) = r_0 - \tau d, \quad r_0 = \hat{w} - c(\tilde{H}^*), \quad r'(d) = -\tau < 0, \quad \hat{r}(d) \xrightarrow{p} r(d) \tag{4}$$

The model delivers additional testable implications beyond the main prediction. First, the treatment effect should be larger for individuals with lower baseline levels of the outcome variable, reflecting diminishing marginal returns. Second, the effect should be persistent over time rather than transitory, because the mechanism operates through permanent changes in the level of the treatment variable. Third, the effect should be concentrated in sectors with greater exposure to the underlying channel. We test each of these auxiliary predictions in Section 6.

We make several simplifying assumptions to maintain tractability. First, we assume that agents have rational expectations and perfect information about the distribution of shocks. Second, we abstract from general equilibrium effects by treating prices as exogenous to

individual agents. Third, we assume that the production technology exhibits constant returns to scale. We discuss the robustness of our theoretical predictions to relaxing each of these assumptions in Appendix A, where we also present a more general version of the model.

The equilibrium of the model can be characterized by a system of first-order conditions and market-clearing conditions. In the interior solution, the marginal benefit of increasing the treatment variable equals the marginal cost, yielding an implicit equation for the optimal level of the treatment variable as a function of exogenous parameters and state variables. We log-linearize this equation around the steady state to obtain the estimating equation that motivates our empirical specification.

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

C Amenity Capitalization

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in urban is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy

applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

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IV Identification Strategy

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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$$\varepsilon_j^S = \varepsilon_0^S - \kappa_1 cdottextUndev_j - kappa_2 cdottextReg_j \tag{5}$$

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variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables

Table 2: IV Estimates: Housing Supply Elasticity (Saiz Land Instrument)

	OLS	IV	IV	IV	First Stage
Second Stage: Dep. var. Δ Log Housing Units					
Δ Log House Price	0.412***	1.841***	1.712***	1.684***	–
	((0.041))	((0.284))	((0.264))	((0.271))	
Δ Log Income	0.184***	0.142**	0.138**	0.131**	–
	((0.038))	((0.058))	((0.061))	((0.064))	
Δ Log Employment	0.218***	0.171***	0.164***	0.158***	–
	((0.032))	((0.048))	((0.051))	((0.054))	
First Stage: Dep. var. Δ Log House Price					
Undevelopable Land Share $\times$ Pop. Growth	–	–	–	–	0.814***
					((0.068))
Slope Index $\times$ Pop. Growth	–	–	–	–	0.641***
					((0.074))
Census Division FE	Yes	Yes	Yes	Yes	Yes
Decade FE	Yes	Yes	Yes	Yes	Yes
Demographic controls	No	No	Yes	Yes	Yes
Regulation controls	No	No	No	Yes	Yes
Observations	954	954	954	954	954
Kleibergen-Paap F-stat	–	84.2	81.4	78.9	–
R-squared (first stage)	–	–	–	–	0.618

Notes: *** p<0.01, ** p<0.05. Instruments: undevelopable land share and terrain slope index (Saiz 2010) interacted with population growth. IV standard errors robust, clustered by MSA.

strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on urban by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

A The Saiz Land Instrument

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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B First Stage and Instrument Validity

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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V Housing Markets and Supply Elasticities

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the

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$$\log P_{ij} = \hat{\alpha}_j + \tilde{\beta} X_{ij} + \varepsilon_{ij}, \quad \hat{\alpha}_j \xrightarrow{p} \alpha_j \text{ as } N_j \to \infty \tag{6}$$

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necessarily approximate but suggest that the magnitudes involved are economically meaningful.

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Table 3: OLS Estimates: Determinants of House Price Growth

	(1)	(2)	(3)	(4)	(5)
Dep. var.: Δ Log Median House Price (decade)					
Δ Log Employment	0.584***	0.512***	0.498***	0.474***	0.461***
	((0.048))	((0.051))	((0.053))	((0.058))	((0.061))
Δ Log Population	0.312***	0.284***	0.271***	0.248***	0.231***
	((0.041))	((0.044))	((0.046))	((0.051))	((0.054))
Δ Log Income	0.241***	0.218***	0.204***	0.191***	0.178***
	((0.038))	((0.041))	((0.043))	((0.047))	((0.050))
College Share Change	–	0.412***	0.398***	0.371***	0.348***
		((0.062))	((0.064))	((0.068))	((0.071))
Amenity Index	–	–	0.184**	0.162**	0.141**
			((0.074))	((0.078))	((0.081))
Supply Elasticity	–	–	–	-0.218***	-0.204***
				((0.044))	((0.048))
Wharton Index	–	–	–	–	0.094**
					((0.038))
Census Division FE	No	Yes	Yes	Yes	Yes
Decade FE	Yes	Yes	Yes	Yes	Yes
1980 baseline controls	No	No	Yes	Yes	Yes
Observations	954	954	954	954	954
R-squared	0.421	0.498	0.531	0.568	0.582

Notes: *** p<0.01, ** p<0.05, * p<0.1. Robust standard errors clustered at MSA level. Sample: 318 MSAs $\times$ 3 decades (1980–1990, 1990–2000, 2000–2010).

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and

Table 4: Housing Supply Elasticity Estimates by City

	Saiz IV	Permit-Based	Hedonic	Combined
Panel A: Constrained Cities (Elasticity $<$ 1.5)				
New York–Northern NJ	0.76	0.82	0.71	0.76
	((0.08))	((0.11))	((0.09))	((0.06))
San Francisco–Oakland	0.64	0.71	0.58	0.64
	((0.09))	((0.12))	((0.10))	((0.07))
Los Angeles–Long Beach	0.84	0.91	0.78	0.84
	((0.08))	((0.11))	((0.09))	((0.06))
Boston	0.91	0.98	0.85	0.91
	((0.09))	((0.12))	((0.10))	((0.07))
Panel B: Elastic Cities (Elasticity $>$ 3.0)				
Houston	5.41	5.18	5.62	5.40
	((0.48))	((0.52))	((0.51))	((0.36))
Phoenix	4.84	4.61	5.02	4.82
	((0.44))	((0.48))	((0.47))	((0.33))
Atlanta	4.12	3.98	4.28	4.13
	((0.38))	((0.42))	((0.41))	((0.29))
Dallas–Fort Worth	4.51	4.34	4.68	4.51
	((0.41))	((0.45))	((0.44))	((0.31))
Mean elasticity (full sample)	2.48	2.41	2.58	2.49
Std. dev.	1.84	1.79	1.91	1.82

Notes: Standard errors in parentheses. Saiz IV: instrument based on undevelopable terrain (Saiz 2010). Permit-based: long-run price elasticity from permit autoregressions. Combined: precision-weighted average.

Table 5: Construction Costs, Land Costs, and the Marginal Cost of Housing

	(1)	(2)	(3)	(4)
Panel A: Dep. var. Log Land Value per Sq. Ft.				
Undevelopable Land Share	1.841***	1.712***	1.584***	1.478***
	((0.184))	((0.192))	((0.201))	((0.214))
Wharton Regulatory Index	0.512***	0.484***	0.454***	0.421***
	((0.068))	((0.071))	((0.074))	((0.078))
Log City Income	0.641***	0.614***	0.581***	0.548***
	((0.071))	((0.074))	((0.078))	((0.081))
Panel B: Dep. var. Log Construction Cost per Sq. Ft.				
Log Material Prices	0.481***	0.452***	0.424***	0.398***
	((0.058))	((0.061))	((0.064))	((0.068))
Log Wage (construction)	0.318***	0.298***	0.281***	0.264***
	((0.041))	((0.044))	((0.046))	((0.049))
Building Height Regulation	0.214***	0.198***	0.184***	0.171***
	((0.031))	((0.033))	((0.035))	((0.037))
State FE	No	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
MSA controls	No	No	Yes	Yes
Interaction terms	No	No	No	Yes
Observations (Panel A)	1,272	1,272	1,272	1,272
Observations (Panel B)	1,272	1,272	1,272	1,272

Notes: *** $p<0.01$. Land and construction cost data from Lincoln Institute of Land Policy. Marginal cost of housing = construction cost + opportunity cost of land.

the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

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A House Price Determinants

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B Supply Elasticity Estimates

Table **??** presents our main estimates. Column (1) reports the baseline specification without controls. Columns (2) through (4) progressively add demographic controls, geographic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results. Following the bounding approach of Oster (2019), we show that the degree of selection on unobservables required to explain away our estimate would need to be substantially larger than the degree of selection on observables.

The estimated coefficient on the treatment variable is statistically significant at the one percent level in all specifications. The magnitude implies that a one standard deviation increase in the treatment variable is associated with a positive and significant of approximately 0.3 standard deviations in the outcome variable. To put this in perspective, this effect size is comparable to the impact of an additional year of schooling on earnings, one of the most well-established causal relationships in economics. The 95 percent confidence interval is sufficiently narrow to rule out effects smaller than one-tenth of a standard deviation.

We find no evidence that our results are driven by pre-existing trends or differential selection. The event study coefficients are small and statistically insignificant in all pre-treatment periods, lending support to our parallel trends assumption. Moreover, the treatment effect emerges sharply at the time of the intervention rather than gradually, which is consistent with a causal interpretation and inconsistent with pre-existing trends that happen to accelerate at the time of treatment.

The first-stage relationship between the proposed instrument and the treatment variable is strong, with an F-statistic well above conventional thresholds for weak instruments. The first-stage coefficient is precisely estimated and has the expected sign. Following the recommendations of Andrews, Stock, and Sun (2019), we also report the effective F-statistic and the tF critical values. Under all weak-instrument diagnostics considered, we

can reject the null of a weak first stage at the 5 percent level.

Heterogeneity analysis reveals that the treatment effect is concentrated among individuals with below-median baseline levels of the outcome variable. This pattern is consistent with the theoretical prediction that the marginal return to the treatment variable is decreasing. We also find suggestive evidence of heterogeneity along demographic dimensions, although the subsample estimates are less precisely estimated due to smaller sample sizes. The difference in treatment effects across subgroups is statistically significant at the 10 percent level.

We complement the regression analysis with non-parametric evidence on the treatment effect. Kernel density estimates of the distribution of the outcome variable show a clear rightward shift for treated individuals relative to untreated individuals. Quantile treatment effects, reported in the appendix, are positive and statistically significant throughout the distribution, with somewhat larger effects in the lower tail. This pattern suggests that the treatment reduces inequality in the outcome variable by disproportionately benefiting those at the bottom of the distribution.

The magnitude of our estimates is informative about the relative importance of the treatment variable as a determinant of the outcome variable. A variance decomposition exercise reveals that variation in the treatment variable can account for approximately 15 to 25 percent of the cross-sectional variance in the outcome variable, depending on the specification. While this leaves substantial residual variation unexplained, it establishes the treatment variable as one of the most quantitatively important determinants identified in the literature.

We also examine the dynamics of the treatment effect by interacting the treatment variable with indicators for different time horizons. The effect is small and statistically insignificant in the first year after treatment, grows steadily over the subsequent three years, and stabilizes at a new long-run level by year four. This gradual emergence of the treatment effect is consistent with the theoretical model, which predicts that the adjustment to

the new equilibrium occurs over multiple periods as agents update their behavior and as general equilibrium effects propagate through the economy.

C Construction and Land Costs

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on urban by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in urban is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

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Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in urban since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to

an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

VI Amenities and Income Sorting

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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navigates between these extremes by exploiting quasi-experimental variation.

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$$widetildeA_j = fracpartiallogr_j/partialA_jpartiallogr_j/partialA_j - partiallogw_j/partialA_jcdotDeltalogr_j \tag{7}$$

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the

Table 6: Amenity Capitalization into House Prices and Wages

	(1) Prices	(2) Prices	(3) Wages	(4) Wages
Dep. var.: Log Median House Price / Log Median Wage				
January Temperature (10-deg F)	0.184***	0.172***	-0.041**	-0.038**
	((0.028))	((0.031))	((0.018))	((0.019))
Coast Dummy	0.214***	0.198***	0.024	0.021
	((0.041))	((0.044))	((0.024))	((0.026))
Log Sunshine Hours	0.148***	0.138***	-0.018	-0.016
	((0.038))	((0.041))	((0.022))	((0.024))
Log Crime Rate	-0.124***	-0.114***	0.028**	0.024*
	((0.024))	((0.026))	((0.014))	((0.015))
School Quality Index	0.094***	0.084***	-0.008	-0.007
	((0.019))	((0.021))	((0.011))	((0.012))
Census Division FE	No	Yes	No	Yes
MSA size controls	Yes	Yes	Yes	Yes
Observations	318	318	318	318
R-squared	0.541	0.584	0.314	0.348

Notes: *** p<0.01, ** p<0.05, * p<0.1. Cross-section, 2000 Census. Amenity values from Albouy (2016) index. Rosen-Roback theory predicts positive price and negative wage amenity gradients.

Table 7: Income Sorting and Skill Sorting Across Cities

	(1)	(2)	(3)	(4)
Dep. var.: Δ Log Median Income or Δ College Share				
Supply Elasticity (Saiz)	-0.048***	-0.041***	-0.081***	-0.074***
	((0.012))	((0.013))	((0.018))	((0.019))
Amenity Index	0.124***	0.114***	0.218***	0.204***
	((0.028))	((0.031))	((0.041))	((0.044))
Initial College Share	0.312***	0.284***	–	–
	((0.041))	((0.044))		
Productivity Growth	0.248***	0.228***	0.412***	0.384***
	((0.038))	((0.041))	((0.058))	((0.062))
Dep. var.	Δ Income	Δ Income	Δ College	Δ College
Census Division FE	No	Yes	No	Yes
Decade FE	Yes	Yes	Yes	Yes
Observations	954	954	954	954
R-squared	0.481	0.524	0.514	0.558

Notes: *** p<0.01. Inelastic supply cities attract higher-income and higher-skill workers as housing constraints intensify income sorting. Supply elasticity coded so higher = more elastic.

channels through which the treatment variable affects the outcome variable.

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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A Amenity Values and Quality of Life

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B Skill Sorting Across Cities

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VII Spatial Equilibrium and Migration

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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$$M_{ij} = exp!left(alpha_i + gamma_j + delta_1 logd_{ij} + delta_2(logw_j - logr_j) + varepsilon_{ij} right) \tag{8}$$

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Table 8: Tests of Spatial Equilibrium: Utility Equalization Across Cities

	(1)	(2)	(3)	(4)
Dep. var.: Net Migration Rate (% per decade)				
Real Wage Advantage (%)	0.618***	0.584***	0.541***	0.512***
	((0.071))	((0.074))	((0.078))	((0.081))
House Price Disadvantage (%)	-0.481***	-0.452***	-0.418***	-0.394***
	((0.058))	((0.061))	((0.064))	((0.068))
Amenity Differential	0.214***	0.198***	0.184***	0.171***
	((0.038))	((0.041))	((0.043))	((0.046))
Unemployment Differential	-0.284***	-0.261***	-0.241***	-0.224***
	((0.044))	((0.047))	((0.050))	((0.053))
Utility Equalization Test (χ^2)	2.84	2.41	2.18	1.98
	([0.241])	([0.299])	([0.336])	([0.372])
Census Division FE	No	Yes	Yes	Yes
Decade FE	Yes	Yes	Yes	Yes
Skill group FE	No	No	Yes	Yes
Race controls	No	No	No	Yes
Observations	954	954	954	954
R-squared	0.468	0.511	0.548	0.572

Notes: *** p<0.01. Utility equalization test: H_0 that coefficients on wage and price differentials sum to zero (spatial indifference). p-values in brackets.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

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Table 9: Migration Gravity Model: Bilateral Flows Between MSAs

	(1)	(2)	(3)	(4)
Dep. var.: Log Bilateral Migration Flow				
Log Distance	-1.284***	-1.218***	-1.184***	-1.148***
	((0.048))	((0.051))	((0.054))	((0.058))
Log Origin Population	0.841***	0.812***	0.784***	0.754***
	((0.031))	((0.033))	((0.035))	((0.038))
Log Destination Population	0.914***	0.884***	0.852***	0.821***
	((0.033))	((0.036))	((0.038))	((0.041))
Destination Wage Premium	0.481***	0.448***	0.412***	0.384***
	((0.064))	((0.068))	((0.071))	((0.074))
Destination Housing Cost	-0.314***	-0.284***	-0.261***	-0.241***
	((0.048))	((0.051))	((0.054))	((0.058))
Common State Dummy	0.684***	0.648***	0.614***	0.584***
	((0.081))	((0.084))	((0.088))	((0.091))
Origin FE	No	Yes	Yes	Yes
Destination FE	No	No	Yes	Yes
Decade FE	Yes	Yes	Yes	Yes
Skill group	No	No	No	Yes
Observations (city pairs)	30,258	30,258	30,258	30,258
R-squared	0.641	0.712	0.748	0.784

Notes: *** p<0.01. Gravity model estimated by PPML. Bilateral flows from Census migration matrices (1990, 2000, 2010). Distance: great-circle between MSA centroids.

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A Testing Utility Equalization

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

B Migration Responses to City Shocks

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on urban by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strat-

egy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in urban is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the

theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

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VIII Land Use Regulation

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on urban by exploiting a novel source of variation in the

treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

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Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable,

depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information

on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Table 10: Land Use Regulation and Housing Supply

	(1)	(2)	(3)	(4)
Dep. var.: Log Annual Housing Permits per Capita				
Wharton Regulatory Index	-0.284***	-0.261***	-0.241***	-0.218***
	((0.038))	((0.041))	((0.044))	((0.048))
Minimum Lot Size (acres)	–	-0.148***	-0.138***	-0.124***
		((0.028))	((0.030))	((0.033))
Zoning Strictness Index	–	–	-0.184***	-0.168***
			((0.034))	((0.037))
Log House Price (lagged)	0.218***	0.198***	0.184***	0.168***
	((0.031))	((0.033))	((0.035))	((0.038))
Undevelopable Land Share	-0.614***	-0.581***	-0.554***	-0.524***
	((0.074))	((0.078))	((0.081))	((0.085))
State FE	No	Yes	Yes	Yes
Decade FE	Yes	Yes	Yes	Yes
MSA size controls	Yes	Yes	Yes	Yes
Income controls	No	No	No	Yes
Observations	954	954	954	954
R-squared	0.514	0.548	0.574	0.598

Notes: *** p<0.01. Wharton Regulatory Index from Gyourko, Saiz, and Summers (2008): higher values indicate more restrictive regulation. Regulation reduces permit issuance conditional on physical constraints.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and

policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on urban by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

A Effects of Regulation on Supply

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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A central question in urban is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

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B Political Economy of Zoning

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this

literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on urban by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

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IX Conclusion

This paper provides new evidence on the causal effect of the treatment variable on the outcome variable. Using an instrumental variables strategy applied to administrative records, we find a positive and significant and statistically significant effect that is robust across a

wide range of specifications. The magnitude of the effect is economically meaningful and has clear implications for both theory and policy.

Our findings have several implications for economic theory. First, the results support models in which the treatment variable plays a central role in determining the outcome variable, as opposed to alternative theories that emphasize other factors. Second, the pattern of heterogeneity in treatment effects is consistent with the predictions of the theoretical framework, providing a rare example of a model that is both internally consistent and empirically validated. Third, the dynamics of the treatment effect shed light on the speed of adjustment to economic shocks, a parameter that is notoriously difficult to estimate.

From a policy perspective, the estimates suggest that interventions targeting the treatment variable could have meaningful effects on the outcome variable. However, several caveats are in order. First, our estimates are local to the population and context studied, and external validity to other settings should not be assumed without additional evidence. Second, our identification strategy relies on assumptions that, while supported by the evidence, cannot be directly tested. Third, the general equilibrium effects of large-scale policy interventions may differ from the partial equilibrium effects captured by our estimates.

Future work should investigate the mechanisms through which the treatment variable affects the outcome variable in greater detail. Our results identify a reduced-form relationship but do not fully pin down the structural channels at work. Complementary evidence from randomized controlled trials, structural estimation, or natural experiments in other contexts would strengthen the conclusions drawn here. Additionally, understanding the long-run persistence of the effects documented in this paper is an important question for future research.

In conclusion, this paper demonstrates that the treatment variable is a quantitatively important determinant of the outcome variable. The combination of a compelling identification strategy, rich data, and robust results across specifications provides a strong foundation for the causal claims made here. We hope that these findings will stimulate further

research on the economic consequences of variation in the treatment variable and will inform the design of policies aimed at improving the outcome variable.

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X Appendix A: Spatial Equilibrium Theory

Consider an economy populated by a continuum of agents indexed by $i \in [0, 1]$. Each agent chooses the outcome variable to maximize lifetime utility subject to a budget constraint. The key source of heterogeneity across agents is their exposure to the treatment variable. We solve for the equilibrium of this economy and derive comparative statics with respect to the key parameters. The model generates a clear testable prediction: an exoge-

nous increase in the treatment variable should lead to a positive and significant change in the outcome variable, with the magnitude depending on structural parameters that we estimate in the empirical analysis.

Let Y_i denote the outcome of interest for unit i. We model Y_i as a function of treatment status D_i, observed covariates X_i, and unobserved heterogeneity ε_i. Under standard regularity conditions, the conditional expectation function takes the form presented in Equation **??**. The parameter of interest is the coefficient on treatment status, which captures the causal effect of the treatment variable on the outcome variable holding constant observed and unobserved confounders. Identification of this parameter requires an exclusion restriction, which we discuss in detail below.

The theoretical framework builds on the canonical model of field. Agents face a discrete choice between J alternatives, each yielding utility $U_{ij} = V_{ij} + \epsilon_{ij}$, where V_{ij} is the deterministic component and ϵ_{ij} captures idiosyncratic preferences. Under standard distributional assumptions on the error term, the choice probabilities take a tractable closed form. The model can be estimated using maximum likelihood or method of moments, and the resulting structural parameters can be used to simulate counterfactual policy scenarios.

Proposition 1 (Existence of Spatial Equilibrium)**.** *Under standard regularity conditions on preferences $U(c, H, A)$ and production, a spatial equilibrium exists. Specifically, suppose: (i) U is continuous, strictly quasi-concave, and strictly increasing in c, H, and A; (ii) the housing supply $S(r; \eta)$ is continuous and increasing in r for all η; (iii) there exist upper and lower bounds on feasible wages and rents. Then there exists at least one equilibrium allocation $(w_j^*, r_j^*, n_j^*)_{j \in \mathcal{J}}$ such that $V(w_j^*, r_j^*, A_j) = \bar{V}$ for all populated cities.*

Proof. Map $T : \Delta \to \Delta$ on the simplex of city population shares. Brouwer's fixed-point theorem applies since T is continuous and Δ is compact and convex. At fixed point n^*, labor market clearing pins wages via $w_j = w_j(n_j^*)$ and housing market clearing pins rents via $r_j = r_j(n_j^*, S_j)$. Utility equalization follows from the definition of T. □

Proposition 2 (Uniqueness with Linear Supply)**.** *If housing supply is log-linear,* $\log H_j =$

$\varepsilon_j^S \log r_j + \zeta_j$*, and preferences are Cobb-Douglas,* $U = c^\alpha H^\beta A^\gamma$ *with* $\alpha + \beta + \gamma = 1$*, then the spatial equilibrium is unique.*

Proof. Log-linearizing around the equilibrium, the system reduces to:

$$d \log r_j = \frac{1}{\varepsilon_j^S} d \log n_j + \frac{1}{\varepsilon_j^S} d \log A_j^{supply}, \tag{9}$$

$$d \log w_j = \frac{\phi}{\varepsilon_j^L} d \log n_j, \tag{10}$$

$$dV = \alpha \, d \log w_j - \beta \, d \log r_j + \gamma \, dA_j = 0, \tag{11}$$

where ε_j^L is the labor demand elasticity. Substituting and solving for dn_j: the Jacobian of this system has a unique zero under the assumption $\varepsilon_j^S > 0$, $\varepsilon_j^L < 0$. □

Proposition 3 (Comparative Statics: Supply Constraint Effect)**.** *Let* $\varepsilon_j^S \to 0$ *(perfectly inelastic supply). Then a positive amenity shock* $dA_j > 0$ *is fully capitalized into rents:* $d \log r_j \to \infty$*, with no change in population* $dn_j = 0$*. When* $\varepsilon_j^S \to \infty$ *(perfectly elastic), the same shock induces unbounded population growth with no rent increase.*

Proof. From the log-linearized equilibrium:

$$d \log r_j = \frac{dA_j + \phi \, dn_j}{\varepsilon_j^S}, \tag{12}$$

$$dn_j = \frac{\varepsilon_j^S \cdot \gamma \, dA_j}{\beta + (\alpha\phi/\varepsilon_j^L - \beta/\varepsilon_j^S) \cdot 0}. \tag{13}$$

Taking $\varepsilon_j^S \to 0$ and $\varepsilon_j^S \to \infty$ establishes the stated limits. □

Lemma 1 (Convergence of Hedonic Estimator)**.** *Let* $\hat{\alpha}_j$ *denote the city fixed effect from the hedonic regression. Then:*

$$\sqrt{N_j} \left(\hat{\alpha}_j - \alpha_j \right) \xrightarrow{d} \mathcal{N}\left(0, \tilde{\sigma}_\varepsilon^2 / \hat{\sigma}_X^2\right), \quad \arg\max_{j \in \mathcal{J}} \hat{A}_j \xrightarrow{p} \arg\max_{j \in \mathcal{J}} A_j. \tag{14}$$

[Amenity Capitalization under Heterogeneous Preferences] When households differ in taste parameter $\tilde{\theta}_i$, the hedonic price gradient satisfies the integral condition:

$$\frac{\partial \log r_j}{\partial A_j} = \int_0^\infty \frac{U_A(\tilde{\theta})}{U_r(\tilde{\theta}) \cdot H(\tilde{\theta})} \, dG(\tilde{\theta}). \tag{15}$$

We derive the key prediction of the model through a simple comparative statics exercise. An exogenous increase in the treatment variable shifts the equilibrium allocation of the outcome variable through two channels: a direct effect operating through prices, and an indirect effect operating through the reallocation of factors across sectors. The total effect is ambiguous in general, but under the empirically relevant parameter values, we show that the direct effect dominates. Proposition 1 in Appendix A formalizes this result and establishes conditions under which the sign of the total effect is unambiguous.

XI Appendix B: Geographic Instrument Validity

This appendix provides additional results referenced in the main text. We organize the supplementary material into three parts. Appendix A presents proofs of the theoretical propositions stated in Section **??**. Appendix B reports additional empirical results, including alternative specifications and sample definitions. Appendix C describes the construction of key variables in detail.

We present detailed derivations of the key theoretical results stated in Section **??**. The proofs proceed under standard regularity conditions that are verified in the data. We state these conditions formally as Assumptions A1 through A4 below, and provide discussion of their economic content and empirical plausibility. Throughout, we maintain the notation introduced in the main text.

This section reports results from additional robustness exercises not included in the main text due to space constraints. All estimates are qualitatively similar to the baseline results reported in Table **??**. We organize these results by the type of robustness check:

alternative control variables, alternative sample restrictions, alternative functional forms, alternative estimation methods, and alternative definitions of key variables.

We provide here a detailed description of the data sources and variable construction procedures. For each variable, we report the original source, the years available, the unit of observation, and any transformations applied. We also report summary statistics for the full set of variables used in the analysis, including those that appear only in robustness checks or auxiliary regressions.

Table 11: Appendix: Geographic Land Availability and Instrument Validity

	Mean	SD	Min	Max
Panel A: Geographic Constraint Measures				
Undevelopable Land Share (%)	19.8	14.2	0.0	71.4
Terrain Slope Index	2.84	2.41	0.12	11.84
Wetlands Share (%)	4.21	6.84	0.0	38.4
Floodplain Share (%)	8.14	9.21	0.0	41.8
Panel B: Instrument Validity Tests				
Correlation with 1900 Population Density	0.081	–	–	–
	([0.142])			
Correlation with 1900 Income	0.064	–	–	–
	([0.248])			
Correlation with 1900 Manufacturing Share	0.048	–	–	–
	([0.384])			
MSAs	318	318	318	318

Notes: Geographic constraint data from Saiz (2010) based on NLCD land cover and USGS slope data. Pre-determined geographic variables uncorrelated with historical economic conditions, supporting instrument exogeneity.

The following tables report the complete set of coefficient estimates for all control variables included in our main specifications. In the interest of space, the main text reports only the coefficients on the key independent variables. The full results presented here confirm that the control variables enter with expected signs and magnitudes, and that their inclusion does not materially affect the coefficients of interest.

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